To the Editors,
International Journal of Heat and Mass Transfer

Dear Editors,

Please consider our manuscript, "The Boussinesq–compressible gap in steady two-dimensional buoyancy-driven enclosure convection is linear to leading order in the relative temperature contrast: a factorised physics prior for surrogates", for publication in the *International Journal of Heat and Mass Transfer* as a full-length article.

Surrogates for thermal convection are trained overwhelmingly on Boussinesq data, yet the flows engineers need are compressible. The two are close in integral quantities such as the Nusselt number and differ in the local fields that surrogates are asked to reproduce. The manuscript establishes what sets the size of that difference and turns the answer into a design rule.

- **One axis.** In the non-dimensionalisation used by the released paired data and by our solver, the Froude number is pinned ($Fr^2 = \varepsilon$ identically) and the Mach number enters at $O(Ma^2)$; a 24-fold change in Ma leaves the measured correction unchanged at the precision of the solve. The break-symmetry group is therefore one-dimensional, spanned by $\varepsilon = \Delta T/T_0$.
- **A measured law.** Across 627 paired two-dimensional solves in the unit square at $Pr = 0.71$ spanning $\varepsilon = 0.02$–1.9, the relative correction obeys $R = A\varepsilon^m$ with $m = 1.02$ in u_x and u_y. The pressure channel gives $m = 1.01$ in a fixed post-processing gauge; it is a numerical observation, not a gauge-invariant exponent. θ is a weak, flow-dependent channel whose free power-law fit is not identifiable ($R^2 = 0.22$; slope 1.02 over the full window, 1.67 on the 104-configuration subset), so we make no exponent claim for it. The intercept vanishes and the ε^2 curvature matches the asymptotic expansion. In 104-sample checks in a 2:1 geometry (0.7–1.8 standard errors) and at $Pr = 4.38$ (1.5%) the u_x/u_y exponent is consistent; the median correction reproduces on an independent public data set at $\varepsilon = 1.2$.
- **A design rule.** Because the correction factorises as $D = \varepsilon P(X)$, a network that predicts only P has a 1.09× lower channel-averaged error than the strongest conditional baseline in extrapolation (θ and p individually, p in the fixed gauge; the u_x/u_y intervals include zero), and 1.87× better (channel-averaged; 1.89× in u_x) than an ε-blind network when ε is withheld, which makes single-contrast public data reusable at other contrasts within one boundary-condition family, without retraining.
- **A consistency check on the linearised operator.** The factorisation is not only a fit. One Newton step of the low-Mach residual at the Boussinesq state, on the same discretisation, reproduces the measured correction to median 4.7/3.8/7.2% ($u_x/u_y/p$) on 27 samples at $\varepsilon = 0.02$–0.10, with no extrapolation in ε; and the extrapolated shape predicts held-out measurements at $\varepsilon = 0.1$ and 0.2 to 2.4/1.8/3.8%.
- **Wall heat transfer.** At $\varepsilon = 1.2$ the integral Nusselt number changes by 0.67% (mean per-sample ratio 0.54%, median 0.58%), while the local wall heat flux is redistributed by 4.25%; the integral agreement therefore conceals a local error that a surrogate must be trained to resolve. A prior trained on the public data cuts the temperature-field error from 7.2% to 0.10% and the local wall-flux error to 4.3%, but its integrated wall-Nu error is 2.6%, against 0.7% for doing nothing: the wall-normal derivative, not the field norm, is the limiting quantity. A wall-flux penalty in the same loss brings the integrated error to 0.11% and the local error to 1.0%.

IJHMT has published numerical methodology for non-Oberbeck–Boussinesq flows (Demou, Frantzis and Grigoriadis, *Int. J. Heat Mass Transfer* 125 (2018) 1156–1168), so the present study fits the journal's scope: it quantifies that effect and turns it into a surrogate design rule.

The main sweep is two-dimensional and steady; a check on twenty-four paired three-dimensional samples at two Rayleigh numbers reproduces the exponent, and a POD-ROM with at most four modes on the fifteen $Ra = 10^4$ samples lowers the mean u_x error from 0.168 to 0.081 over five held-out contrasts. We state the remaining limits (one resolution, $res = 16$, $\varepsilon \leq 1.9$) explicitly. Every number in the paper is regenerated from stored solver output by the scripts in the repository; the public data set of the companion study is used unchanged, and our scripts and machine-readable results are released with the manuscript; the raw solver output is available from the authors on request.

The work is not under consideration elsewhere, and the authors declare no competing interests.

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